

Islam Asous

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EDUCATION

Hunter College, City University of New York (CUNY)

New York, NY

Bachelor of Arts in Computer Science, Minor in Mathematics

expected Dec 2026

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Computer Theory, Database Management, Object-Oriented Programming, Programming Methods, Advanced AI & ML Theory, Natural Language Processing, Automation Engineering, Web Development, iOS Development, Linear Algebra, Applied Statistics

EXPERIENCE

Project Sisypheus, Clemson University

June 2026 – Present

Software Engineer REU

Remote

- Debugged and maintained an authentication-tracking browser extension (**WXT**, **Vue 3**, **TypeScript**, **Dexie.js**) supporting NSF-funded research (Award #2521037, \$650K) into usability gaps and authentication-fatigue-driven password reuse, capturing real-world authentication metrics across enterprise and AI-crowdsourcing platforms.
- Built a real-time broadcast notification system from scratch (**Vue 3**, **Pinia**, **Supabase Realtime**) enabling admins to push instant announcements to authenticated dashboard users, enforcing role-based access control via Postgres Row-Level Security.
- Embedded **Claude Code** into an agentic full-stack development workflow spanning implementation, test authoring, cross-repository debugging (Supabase permissions, OAuth redirects, CI issues), and PR review: iterating on prompts and AI-generated output to accelerate feature delivery while maintaining 90%+ test coverage.

CUNY Tech Prep

May 2026 – Present

Data Science Fellow

New York, NY

- Selected as 1 of 200 Fellows for the competitive, year-long CUNY Tech Prep Data Science track placing top CUNY students into technical roles.
- Advancing proficiency in Python, machine learning, SQL, and end-to-end data analysis through a project-based curriculum.

Hunter College Computer Science Department

Aug 2025 – Present

Undergraduate Teaching Assistant

New York, NY

- Supported 600+ students across Computer Architecture I & II, Computer Theory, and OOP/Data Structures & Algorithms by leading weekly review sessions and office hours.
- Co-designed quizzes, midterms, and finals for courses of 150–300+ students; reduced grading turnaround by building automated evaluation tools with faculty.

PROJECTS

AI Agent From Scratch, Tool-Calling Research Agent | *Python, LangChain, Anthropic Claude API, Pydantic*

- Engineered a tool-calling research agent using LangChain and the Claude API that autonomously decides whether to invoke a live web-search tool or answer directly based on the query.
- Built a structured-output layer with Pydantic to parse raw model responses into a validated schema (topic, summary, sources, tools used) diagnosing a prompt-adherence failure and iterating on system instructions to enforce reliable, parser-compatible output across both tool-invoking and direct-answer queries.

RAG From Scratch, Local RAG Pipeline | *Python, Ollama, Llama 3.2, BGE Embeddings, Streamlit*

- Engineered a retrieval-augmented generation pipeline from first principles, implementing custom embedding storage, cosine similarity search, and prompt construction without relying on frameworks like LangChain.
- Built an in-memory vector database using a locally-hosted BGE embedding model and developed a Streamlit chat interface with multi-turn conversations, document uploads, and an inspectable retrieved-context view by iterating on prompt design to improve answer grounding.

For the Culture | *React, Vite, Express.js, Google Generative AI, Firebase*

- Expanded access to real-time food, health, and transit resources across all 5 boroughs in 8 languages by co-building and deploying a full-stack multilingual platform with a 4-person team, integrating a generative AI model for on-demand translation.
- Owned the React/Vite frontend architecture, including an interactive map consuming live NYC Open Data REST endpoints with real-time filtering by borough and resource type.

Q69 Real-Time Bus Tracker & SMS Alert System | *Python, MTA BusTime API, Twilio*

- Integrated the MTA BusTime SIRI API to fetch and parse live bus arrival data from nested JSON responses for a specific transit stop.
- Automated end-to-end transit monitoring via a cron-scheduled pipeline that queries live API data, processes arrival times, and dispatches SMS alerts via Twilio's REST API on a set interval thus requiring zero manual intervention.

LEADERSHIP

Math Association of Hunter College

Aug 2024 – Aug 2026

President

New York, NY

- Led the club's officer board, planning and running member events and meetings for the Hunter College math community.

TECHNICAL SKILLS

Programming: Python3, TypeScript, JavaScript, C++, Rust, Swift/SwiftUI, Dart, HTML/CSS, SQL (PostgreSQL)

Frameworks, Cloud & Tools: React, Node.js, Express.js, Flutter, Firebase, Supabase, Docker, REST APIs, Git/GitHub

Certifications: AWS Skill Builder - Developer, Cloud Operator (SysOps), DevOps (In Progress); CodePath AI201: Applications of AI Engineering (In Progress); Databricks Generative AI Fundamentals (In Progress)